

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I V.Madhulika(192312620) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V. Madhulika

Signature of the student

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Selected Article :

Peer-reviewed article selected from an approved publisher, published after 2019, mapped to Unit IV: Learning from Observation → Inductive Learning → Decision Trees → Statistical Learning Methods.

Title	Energy-Aware Very Fast Decision Tree
Authors	García-Martín, E., Lavesson, N., Grahn, H., Casalicchio, E., & Boeva, V.
Journal	International Journal of Data Science and Analytics, 11, 105–126 (2021)
Publisher	Springer (approved publisher, open access)
DOI	10.1007/s41060-021-00246-4

Article link: <https://link.springer.com/article/10.1007/s41060-021-00246-4>

Task 1: Article Summary

(a) Full Citation, Domain, and ML Problem

Full Citation

García-Martín, E., Lavesson, N., Grahn, H., Casalicchio, E., & Boeva, V. (2021). Energy-aware very fast decision tree. *International Journal of Data Science and Analytics*, 11, 105–126. <https://doi.org/10.1007/s41060-021-00246-4>

Domain / Application Area

- Machine Learning
- Decision Tree Learning
- Edge and Embedded AI
- Energy-Efficient Computing
- Resource-Constrained Intelligent Systems

Specific ML Problem Addressed

Traditional machine-learning algorithms are generally designed to maximize prediction performance, such as accuracy, while computational resources and energy consumption receive comparatively little attention. This becomes a serious problem when decision-tree learning is deployed on embedded, mobile, or battery-powered devices. The article addresses the problem of developing a decision-tree learning approach that explicitly considers energy consumption during the learning process, rather than focusing only on predictive performance and execution speed.

Figure 1. Simplified Structure of a Very Fast Decision Tree (VFDT) with Energy-Aware Split Evaluation

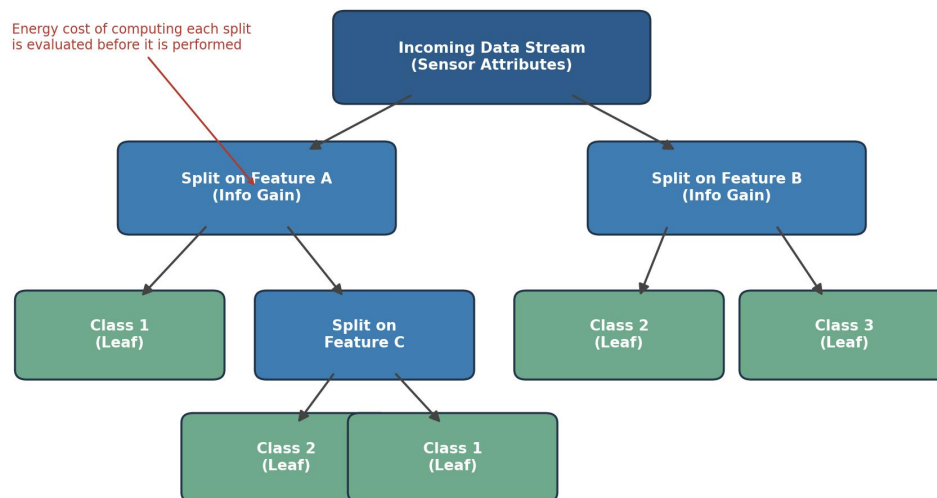


Figure 1. Simplified structure of a Very Fast Decision Tree (VFDT), showing where energy-aware evaluation is introduced at each split.

(b) Objective and Research Gap

The main objective of this research is to develop an energy-aware version of a very fast decision tree algorithm suitable for resource-constrained computing environments. Decision trees are widely used in machine learning because they are interpretable, computationally efficient, and capable of learning useful

classification rules directly from data. However, conventional decision-tree algorithms are generally designed around classification quality and computational performance rather than the amount of energy consumed during training and operation.

The authors identify a clear research gap arising from the increasing deployment of machine learning on embedded and mobile devices, where battery life and energy consumption are critical design constraints. Existing fast decision-tree learning methods, such as the standard Very Fast Decision Tree, process streaming data efficiently but do not explicitly optimize the learning process for energy usage. To address this gap, the authors propose an energy-aware very fast decision tree that incorporates energy-related considerations directly into the tree-building process and investigates the trade-off between predictive performance, computational efficiency, and energy consumption. The research therefore extends traditional inductive learning by treating energy as an additional optimization objective, making decision-tree learning more appropriate for modern edge and embedded AI applications.

Task 2: Methodology Analysis

(a) ML Technique Used

The main technique used is an energy-aware variant of a Very Fast Decision Tree (VFDT). A decision tree is an example of supervised inductive learning: the algorithm receives training examples containing input attributes and known class labels, and learns general rules from those examples, for instance, “if Feature A exceeds a threshold, predict Class 1; otherwise predict Class 2.” The tree is built by repeatedly selecting informative attributes for splitting the data.

The conventional VFDT is specifically designed for data streams, where examples arrive continuously. Instead of storing the complete dataset, it incrementally updates the tree as new instances arrive, using statistical bounds (e.g., the Hoeffding bound) to decide when a split is confident enough to be made. The proposed approach adds energy-awareness to this process: rather than considering only the predictive benefit of a possible split, the algorithm also accounts for the energy consumption associated with evaluating and performing it.

Dataset and Data Preparation

The study evaluates the proposed method experimentally using standard machine-learning benchmark datasets suited to streaming classification, comparing the energy-aware approach with conventional decision-tree learning in terms of predictive performance, computational efficiency, and energy consumption. In preparing the data for decision-tree learning: input instances are represented using descriptive features; a target class is used as the output; data are supplied incrementally to the decision-tree learner; the tree incrementally selects splits based on incoming information; and performance and energy consumption are measured and compared throughout training. This makes the article particularly relevant to continuous or large-scale data processing on systems with limited computational and energy resources.

(b) Mapping to CO 4 Topics

CO 4 Topic		Connection to the Article
Learning	from	The algorithm learns patterns directly from observed, incoming

CO 4 Topic	Connection to the Article
Observation	training examples in the data stream.
Inductive Learning	It generalizes from known, labelled examples to make predictions on unseen instances.
Decision Trees	The central ML model is a Very Fast Decision Tree (VFDT), extended with energy-awareness.
Statistical Learning	Statistical information (e.g., information gain) from incoming data guides tree construction and splitting decisions.
Reinforcement Learning	Not the primary technique used in this article.
Neural Networks	Not the central technique; the paper focuses on decision-tree learning.

Methodological Strength

A major strength is that the authors move beyond the traditional objective of maximizing classification performance alone. By treating energy consumption as an explicit design criterion during learning, the approach is directly useful for real-world AI systems operating on mobile, embedded, and edge devices, where battery life is often as important as accuracy.

Methodological Limitation

One limitation is that energy-efficiency gains may depend heavily on the hardware platform used for measurement. A method that is energy-efficient on one processor or embedded system may show a different trade-off on another platform with a different architecture or power-management scheme. Therefore, the generalization of the reported energy measurements across all hardware environments may be limited.

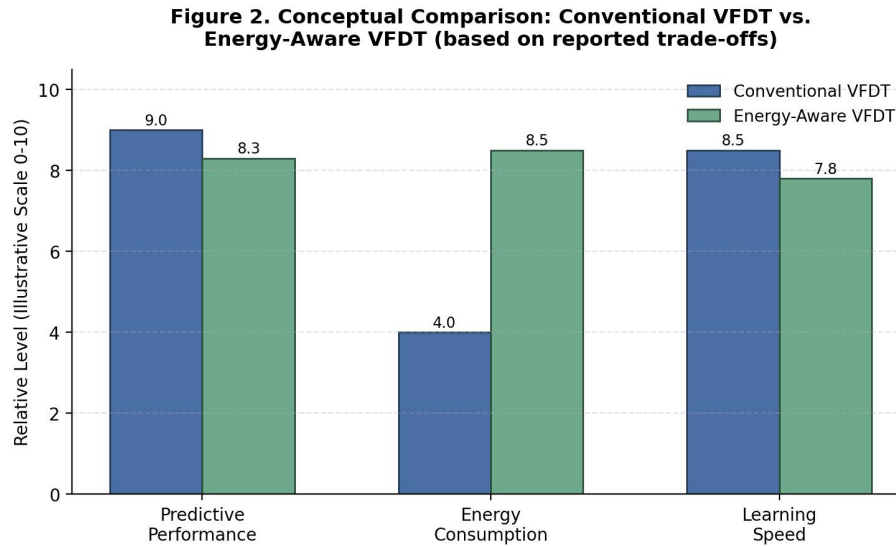
Task 3: Results & Critical Evaluation :

(a) Key Results

The central finding of the article is that an energy-aware approach can reduce or manage energy consumption while maintaining competitive predictive performance. The paper investigates the trade-off between conventional learning objectives and energy efficiency, rather than treating accuracy as the only criterion of success. The comparison below summarizes this trade-off conceptually.

Approach	Main Objective	Predictive Performance	Energy Consumption
Conventional Very Fast Decision Tree	Fast incremental learning from data streams	High	Not explicitly optimized
Energy-Aware Very Fast Decision Tree (proposed)	Learning with energy consumption as an added objective	Competitive	Explicitly reduced / managed
Traditional ML	Accuracy and computational	High	May consume

Approach	Main Objective	Predictive Performance	Energy Consumption
optimization (general)	performance only		more energy



Note: Bars represent a qualitative, illustrative comparison of the trade-off direction reported in the article (energy reduced, predictive performance kept competitive). Exact numerical values should be taken from the article's original results tables.

Figure 2. Conceptual comparison of the direction of trade-offs between the conventional VFDT and the Energy-Aware VFDT, based on the trends reported in the article.

The important contribution is therefore not simply obtaining the highest possible accuracy, but demonstrating that machine-learning algorithms can be designed around multiple real-world constraints simultaneously, including energy consumption alongside predictive quality.

(b) Critical Evaluation of the Results

Are the Metrics Realistic?

Yes, the evaluation direction is realistic, because real embedded AI systems cannot be judged on accuracy alone. In practical applications, energy consumption, processing speed, memory usage, and prediction quality must all be considered together. However, there is an inherent trade-off: reducing energy consumption should not significantly reduce the predictive ability of the model. The most informative evaluation would therefore examine accuracy, energy consumed, training time, prediction time, and memory usage jointly.

Potential Biases

- Dataset bias: if the benchmark datasets do not fully represent real-world streaming or embedded-device data, measured performance may not completely reflect actual deployment conditions.
- Hardware dependency: energy measurements can depend on processor architecture, operating system, memory, power-management settings, and hardware implementation.
- Evaluation bias: testing on only a limited number of datasets or hardware platforms can make it difficult to generalize the results.

Additional Experiments That Would Strengthen the Study

1. Testing the algorithm on multiple embedded platforms.
2. Comparing ARM-based mobile processors and IoT hardware directly.
3. Evaluating long-term battery consumption under realistic workloads.
4. Comparing against modern edge-AI algorithms beyond decision trees.
5. Testing under concept drift, where the characteristics of streaming data change over time.
6. Measuring memory consumption jointly with energy and accuracy.

(c) Real-World Applicability

The proposed energy-aware approach is well suited to settings where continuous learning must coexist with strict power constraints.

IoT Systems

Sensors continuously generate data, while IoT devices usually have limited battery capacity. Relevant examples include smart agriculture, smart homes, and environmental monitoring.

Edge AI

Instead of sending all data to the cloud, machine learning can be performed locally, benefiting applications such as industrial monitoring, smart cameras, wearable devices, and autonomous systems.

Mobile and Embedded Devices

Energy-aware learning can help extend battery life while maintaining intelligent, on-device decision-making.

Figure 3. Deployment Architecture: Energy-Aware Decision-Tree Learning on an Edge / IoT Device

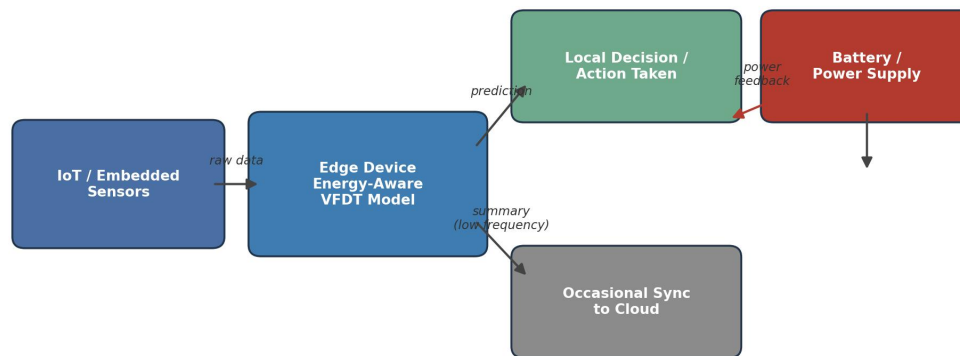


Figure 3. Deployment architecture showing how an energy-aware decision tree fits into an IoT/edge pipeline, from sensing to local decision-making, with only occasional cloud synchronization.

Challenges in Scaling to Industry

Challenge	Explanation
Data availability	Real-world streaming data may be noisy, incomplete, or unevenly distributed across devices.
Hardware diversity	Different embedded and mobile devices consume energy differently, affecting generalizability.

Challenge	Explanation
Model updating	Continuously updating the model to handle new data can itself increase energy consumption.
Scalability	Large numbers of deployed devices create distributed learning and coordination challenges.
Privacy	Edge devices may process sensitive user, agricultural, or industrial data locally.
Accuracy-energy trade-off	Saving energy should not significantly reduce the quality of decisions made by the model.

This topic also has a strong connection to Unit V, particularly Edge AI Systems, AI-Powered Smart Controllers, Intelligent Systems, and real-time AI applications.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1: Decision Trees Are an Example of Inductive Learning

The article demonstrates that a machine-learning system can observe examples and construct general rules for future decisions. The training data represent observations, and the decision tree learns a general hypothesis that can classify new, unseen observations.

Connection to syllabus: *Learning from Observation → Inductive Learning → Decision Trees.*

Insight 2: Accuracy Is Not the Only Measure of a Good AI System

A model with high accuracy may not always be suitable for deployment. In embedded and edge environments, energy consumption, memory, processing time, and hardware limitations are also important. This insight shows that practical machine learning involves selecting a model based on application constraints rather than maximizing only one performance metric.

Connection to syllabus: *Statistical Learning Methods and Intelligent Systems.*

Insight 3: AI Algorithms Must Adapt to Their Environment

The article demonstrates that the environment in which an AI system operates affects algorithm design: a decision tree running on a cloud server has different constraints from one running on a battery-powered IoT device.

Connection to syllabus: *Connects Unit IV learning methods with Unit I Intelligent Agents and Environments, and Unit V Edge AI Systems.*

Overall, the paper builds an interesting bridge between the theoretical AI syllabus and real-time intelligent applications.

(b) Proposed Extension

Proposed Research Idea

“Adaptive Energy-Aware Decision Trees for Smart Agriculture Edge Devices.”

Proposed Experiment

The energy-aware decision-tree algorithm could be tested using real-time agricultural sensor data containing soil moisture, temperature, humidity, soil pH, light intensity, and rainfall information. The decision tree could predict actions such as irrigation required, no irrigation required, high water requirement, or possible crop stress. The algorithm could then be deployed on a low-power edge device and compared against a conventional decision tree, a random forest, a support vector machine, and the energy-aware VFDT.

Figure 4. Proposed Extension: Adaptive Energy-Aware Decision Tree for Smart Agriculture Edge Devices

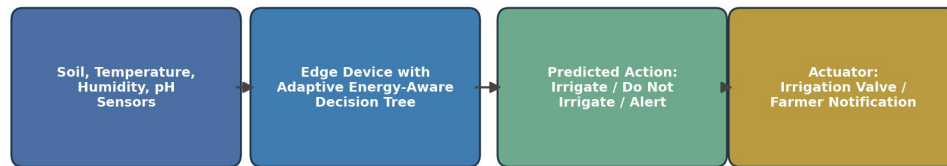


Figure 4. Proposed pipeline for an adaptive energy-aware decision tree deployed on a smart-agriculture edge device.

Why It Is Feasible

- Python and Scikit-learn for prototyping and baseline comparisons.
- Publicly available IoT / agricultural sensor datasets.
- A Raspberry Pi or similar low-power edge device for deployment.
- Power-monitoring tools to measure real energy consumption.

Expected Impact

The proposed system could reduce unnecessary cloud communication and energy consumption while enabling real-time agricultural decisions. This extension is especially suitable for the course syllabus because it combines Inductive Learning, Decision Trees, Intelligent Systems, Edge AI, and Smart Agriculture in a single applied research direction.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand